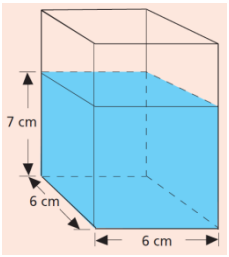
Exercise

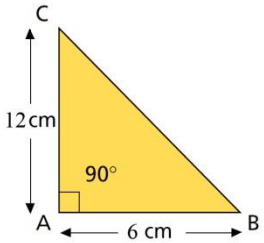
I. Chapter1: Lesson1 Measurement of mass, length and time

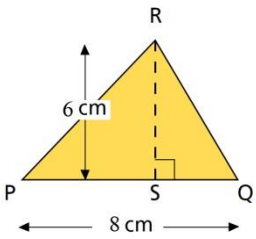
1. How many basic quantities in physics? List its down and express their units.
2. What is the definition of power of tens shorthand? Give example at least three.
3. What is the definition of volume, mass, time, and area?
4. How to convert unit of submultiple of length from large to small and from small to large?
5. How many millimetres are there in
   a 1 cm, **b** 4 cm, **c** 0.5 cm, **d** 6.7 cm, **e** 1 m?
6. What are these lengths in metres:
   a 300 cm, **b** 550 cm, **c** 870 cm, **d** 43 cm, **e** 100 mm?
7. How many micrometers are there in
   a 1cm, **b** 4cm, **c** 0.5cm, **d** 6.7cm, **e** 1m?
8. How many nanometers are there in
   a 1cm, **b** 4cm, **c** 0.5cm, **d** 6.7cm, **e** 1m?
9. Consider the units of volume as following then convert these units to cubic centimeter (cm^3).
   a $20m^3$, **b** $0.2m^3$, **c** $200m^3$, **d** $6.7m^3$, **e** $20.5m^3$?
10. Consider the units of volume as following then convert these units to cubic millimeters (mm^3).
    a $20m^3$, **b** $0.2m^3$, **c** $200m^3$, **d** $6.7m^3$, **e** $20.5m^3$?
11. Consider the units of volume as following then convert these units to cubic decimeter (dm^3).
    a $40m^3$, **b** $0.4m^3$, **c** $600m^3$, **d** $8.7m^3$, **e** $20.5m^3$?
12. Consider the units of volume as following then convert these units to cubic centimeter (cm^3).
    a 40ml, **b** 0.4ml, **c** 600ml, **d** 8.7ml, **e** 20.5ml?
13. Consider the units of volume as following then convert these units to cubic meter (m^3).
    a 40ml, **b** 0.4ml, **c** 600ml, **d** 8.7ml, **e** 20.5ml?
14. Consider the units of area as following then convert these units to meters square (m^2).
    a $29dm^2$, **b** $0.29\ dm^2$, **c** $2900\ cm^2$, **d** $2.9\ cm^2$, **e** $20.5\ mm^2$?
15. Consider the units of area as following then convert these units to millimeter square (mm^2).
    a $29m^2$, **b** $0.29\ m^2$, **c** $2900\ dm^2$, **d** $2.9\ dm^2$, **e** $20.5\ cm^2$?
16. **a** Write the following as powers of ten with one figure before the decimal point:
    100 000 3500 428 000 000 504 27 056
    b Write out the following in full:
    10^3 2×10^6 6.92×10^4 1.34×10^2 10^9
17. **a** Write these fractions as powers of ten:
    1/1000 7/100 000 1/10 000 000 3/60 000
    b Express the following decimals as powers of ten with one figure before the decimal point:
    0.5 0.084 0.000 36 0.001 04
18. The pages of a book are numbered 1 to 200 and each leaf is 0.10 mm thick. If each cover is 0.20 mm thick, what is the thickness of the book?
19. How many significant figures are there in a length measurement of:
    a 2.5 cm, **b** 5.32 cm, **c** 7.180 cm, **d** 0.042 cm?
20. A rectangular block measures 4.1 cm by 2.8 cm by 2.1 cm. Calculate its volume giving your answer to an appropriate number of significant figures.
21. A metal block measures 10 cm × 2 cm × 2 cm. What is its volume? How many blocks each 2 cm × 2 cm × 2 cm have the same total volume?
22. How many blocks of ice cream each 10 cm × 10 cm × 4 cm can be stored in the compartment of a freezer measuring 40 cm × 40 cm × 20 cm?
23. A Perspex container has a 6 cm square base and contains water to a height of 7 cm (Figure 1.13).
    a What is the volume of the water?
    b A stone is lowered into the water so as to be completely covered and the water rises to a height of 9 cm. What is the volume of the stone?



24. **a** Calculate the area of triangle as shown in figure at the right hand side.

b. A rectangular block of wood has dimensions 240 mm × 20.5 cm × 0.040cm. Calculate its volume in cm^3.





II. Chapter1 lesson2 Density

25. **a** If the density of wood is 0.5 g/cm^3 what is the mass of
(i) 1 cm^3,
(ii) 2 cm^3,
(iii) 10 cm^3?
b What is the density of a substance of
(i) mass 100 g and volume 10 cm^3,
(ii) volume 3cm^3 and mass 9 kg?
c The density of gold is 19 g/cm^3. Find the volume of
(i) 38 g,
(ii) 95 g of gold.

26. A piece of steel has a volume of 12 cm^3 and a mass of 96 g. What is its density in
a g/cm^3,
b kg/cm^3?

27. What is the mass of 5 cm^3 of cement of density 3000 kg/cm^3?

28. What is the mass of air in a room measuring 10 m × 5.0 m × 2.0 m if the density of air is 1.3 kg/cm^3?

29. When a golf ball is lowered into a measuring cylinder of water, the water level rises by 30 cm^3 when the ball is completely submerged. If the ball weighs 33g in air, find its density.

30. Why does ice float on water?

31. The volume of a rectangular solid, with a length of 1.5 cm, is 10.98 cm^3. The width of the same solid is 1.2 cm. Find its height.

32. A cube has length of 7 cm a width of 8 cm and a height of 2 cm. What is its volume? If the cube has a mass of 50 grams, what is its density?

33. A marble has a volume of 20 ml. It also has a mass of 10g. Calculate the density of the marble.

34. A box has a length of 5 cm, width of 10 cm and a height of 2 cm. If the box has a mass of 20g, what is the density of the box?

35. The density of aluminum is 2.7 g/ml. If the aluminum has a volume of 20 ml, what is the mass of the aluminum?

36. A rock has a density of 8 g/ml. If the rock's mass is 2g, what is the rock's volume?

37. The length and width of a rectangular solid are 2 cm. Its volume is 36 cm^3. Find the height of this rectangular solid.

38. What is the density of a steel nail whose volume is 3 cm^3 and whose mass is 24 g?

39. Find the mass of a cork whose density is 0.12 g/cm^3 and whose volume is 9.0cm^3?

40. An ice cube's volume is 4.9cm^3. Find its mass if its density is 0.92 g/cm^3.

41. A solid plastic ball's mass is 225g. The density of the plastic is 2 g/cm^3. What is the volume of the ball?

42. Find the volume of an ice cube whose mass is 2 g. The density of ice is 0.92g/cm^3.

43. An object has a mass of 57.7g and occupies a volume of 21.65cm^3, what is its density?

44. An object whose density is 1.98 x 10^3 kg/m^3 has a mass of 35.4g. How many cubic centimeters does the object occupy?

45. A cylinder has a radius of 12.0 cm. It has a density of 15.2 g/cm^3 and a mass of 100g. What is the height of the cylinder? (Hint: volume of a cylinder can be calculated by $\pi r^2 h$)

46. A piece of metal has a mass of 90.51g and its volume is 11.5cm^3. What is the density of the metal?

47. Find the density of a gold bar that has a mass of 1,930g and a volume of 100 cm^3.

48. What is the density of a bar of soap that has a volume of 80 cm^3 and a mass of 90 g?

49. A rectangular block of stone has a length of 12.3cm, a width of 7.6cm, and a height of 4.7cm. What is the volume of the stone block?

50. What is the volume of a brick that is 20.3cm long, 8.9cm wide, and 5.7cm high?

51. Find the volume of a box with a height of 15cm, a width of 18cm, and a length of 30cm.

52. Lead has a density of 11.3 g/cm3. If a piece of lead has a volume of 4cm^3, what is its mass?

53. A stainless steel rod has a mass of 59.2 g and a density of 7.9 g/cm^3. What is the volume of the rod?

III. Chapter2 lesson1 Thermometer

54. How many temperature scales in physics? list its down.

55. What is the clinical thermometer?

56. What is different between temperature and heat?

57. 1530℃ 120℃ 55℃ 37 ℃ 19℃ 0℃ −12℃ −50℃

From the above list of temperatures choose the most likely value for *each* of the following:

a the melting point of iron,

b the temperature of a room that is comfortably warm,

c the melting point of pure ice at normal pressure,

d the lowest outdoor temperature recorded in London in winter,

e the normal body temperature of a healthy person.

58. In order to make a mercury thermometer that will measure small changes in temperature accurately, would you

A decrease the volume of the mercury bulb

B put the degree markings further apart

C decrease the diameter of the capillary tube

D put the degree markings closer together

E leave the capillary tube open to the air?

59. **a** How must a property behave to measure temperature?

b Name three properties that qualify.

c Name a suitable thermometer for measuring

(i) a steady temperature of 1000℃,

(ii) the changing temperature of a small object,

(iii) a winter temperature at the North Pole.

60. Convert 30℃ to Fahrenheit.

61. How many degrees Fahrenheit is 42℃?

62. How many degrees Fahrenheit is 94℃?

63. How many degrees Fahrenheit is 63℃?

64. How many degrees Fahrenheit is t℃?

65. Convert the temperature from Celsius to Kelvin

Example: Convert 33℃ to K

$K = T_C + 273 = 33 + 273 = 306K$

1)	40℃=.....K	2)	82℃=.....K	3)	63℃=.....K
4)	34℃=.....K	5)	-25℃=.....K	6)	91℃=.....K
7)	59℃=.....K	8)	-48℃=.....K	9)	114℃=.....K
10)	66℃=.....K	11)	-7℃=.....K	12)	110℃=.....K

66. Convert the temperature from Kelvin to Celsius

Example: Convert 404K to ℃

$T_C = K - 273 = 404 - 273 = 131℃$

1)	442K=.....℃	2)	241K=.....℃	3)	310K=.....℃
4)	324K=.....℃	5)	428K=.....℃	6)	265K=.....℃
7)	250K=.....℃	8)	371K=.....℃	9)	335K=.....℃
10)	410K=.....℃	11)	283K=.....℃	12)	336K=.....℃

67. Convert the temperature from Celsius to Fahrenheit

Example: Convert 36℃ to °F

$T_F = \frac{9}{5}T_C + 32 = 1.8 \times 36 + 32 = 64.8 + 32 = 96.8°F$

1)	25℃=.....°F	2)	6℃=.....°F	3)	10℃=.....°F
4)	-40℃=.....°F	5)	-5℃=.....°F	6)	103℃=.....°F
7)	30℃=.....°F	8)	-12℃=.....°F	9)	63℃=.....°F
10)	75℃=.....°F	11)	-81℃=.....°F	12)	32℃=.....°F
13)	-212℃=.....°F	14)	94℃=.....°F	15)	-50℃=.....°F

68. Convert the temperature from Fahrenheit to Celsius

Example: Convert 52°F to ℃

$T_C = \frac{5}{9}(T_F - 32) = \frac{5}{9}(52 - 32) = 11.1℃$

1) 40°F=.....℃	2) 168°F=.....℃	3) -200°F=.....℃
4) 63°F=.....℃	5) 82°F=.....℃	6) 178°F=.....℃
7) -114°F=.....℃	8) -76°F=.....℃	9) 59°F=.....℃
10) -404°F=.....℃	11) 122°F=.....℃	12) 185°F=.....℃
13) 97°F=.....℃	14) 211°F=.....℃	15) -50°F=.....℃

IV. Chapter2 lesson2 expansion of solid, liquid

69. Explain why

a the metal lid on a glass jam jar can be unscrewed easily if the jar is inverted for a few seconds with the *lid* in very hot water,

b furniture may creak at night after a warm day,

c concrete roads are laid in sections with pitch between them.

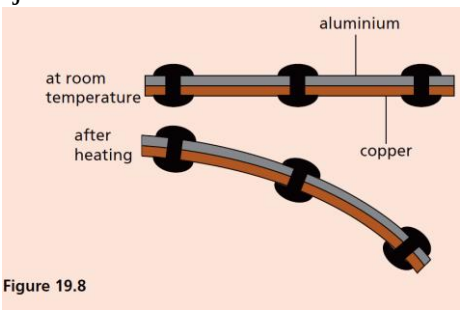


Figure 19.8

70. A bimetallic strip is made from aluminium and copper. When heated it bends in the direction shown in Figure 19.8. Which metal expands more for the same rise in temperature, aluminium or copper? Draw a diagram to show how the bimetallic strip would appear if it were cooled to below room temperature.

71. When a metal bar is heated the increase in length is greater if

1 the bar is long

2 the temperature rise is large

3 the bar has a large diameter.

Which statement(s) is (are) correct?

A 1, 2, 3 **B** 1, 2 **C** 2, 3 **D** 1 **E** 3

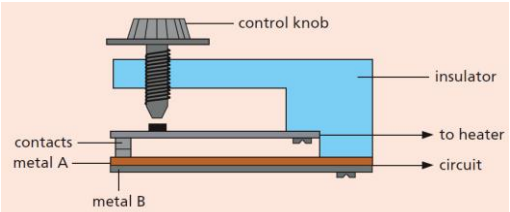


Figure 19.9

72. A bimetallic thermostat for use in an iron is shown in Figure 19.9.

1 It operates by the bimetallic strip bending away from the contact.

2 Metal A has a greater expansivity than metal B.

3 Screwing in the control knob raises the temperature at which the contacts open.

Which statement(s) is (are) correct?

A 1, 2, 3 **B** 1, 2 **C** 2, 3 **D** 1 **E** 3

73. A segment of steel railroad track has a length of 30.000 m when the temperature is 0.0℃.

(A) What is its length when the temperature is 40.0℃? **Answer:** $L_f = 30.013m$

(B) What is its length when the temperature is 80. 0℃? **Answer:** $L_f = 30.026m$

74. The concrete sections of a certain superhighway are designed to have a length of 25.0 m. The sections are poured and cured at 10.08℃. What minimum spacing should the engineer leave between the sections to eliminate buckling if the concrete is to reach a temperature of 50.08℃? Assume that the linear expansion is $12.0 \times 10^{-6}/℃$. **Answer:** $\Delta L = 1.20cm$